# Weekly Update, 2–6 Nov 2026

Brian Zhou · CSE 5999 hiking-speed project · for Danni Beaulieu · 6 Nov 2026

This week all five Gate 2 checks passed, I hand-checked the stop rule, and I tested aerial imagery. Ready-made imagery embeddings (Track A) did not carry over to new regions, so photos centred on the trail (Track B) are now running.

## This week at a glance

| Mon 2 Nov | Wed 4 Nov | Thu 5 Nov | Fri 6 Nov |
|---|---|---|---|
| **Architecture review** learned Tobler's setup, explained to Danni | **Gate 2 brief** five checks and the imagery plan | **Stop-rule check** 20 GPS moments labeled by hand | **Track A tested** helps in known places, not in new regions |
| | | **Clay imagery** 754 tiles checked, 6 GB downloaded | **Track B started** 164,415 trail photos through Satlas |

## Gate 2: all five checks pass

| Gate 2 check | Result |
|---|---|
| Every learned model beats Tobler on held-out hikes | Yes: Tobler's fixed numbers are the worst model in every test |
| Trail tags beat slope alone | Yes: +1.3–1.5% on new hikes |
| Remote sensing compared with tags, gap measured | In unseen regions terrain adds 2.1–2.8%, tags 0.4–0.9%; terrain wins in 16 of 17 regions |
| Uncertainty is calibrated | The 90% ranges hold 89.4–89.9% of test segments |
| Our model beats boosted trees on the same inputs | A tie with the same spread rule; +0.55–0.66 points with its own spread |

**In new regions, learned Tobler is 3.4% better than slope alone; Tobler is 12.5% worse**

CRPS skill vs a slope-only curve, each of 17 regions held out in turn; best version of each model



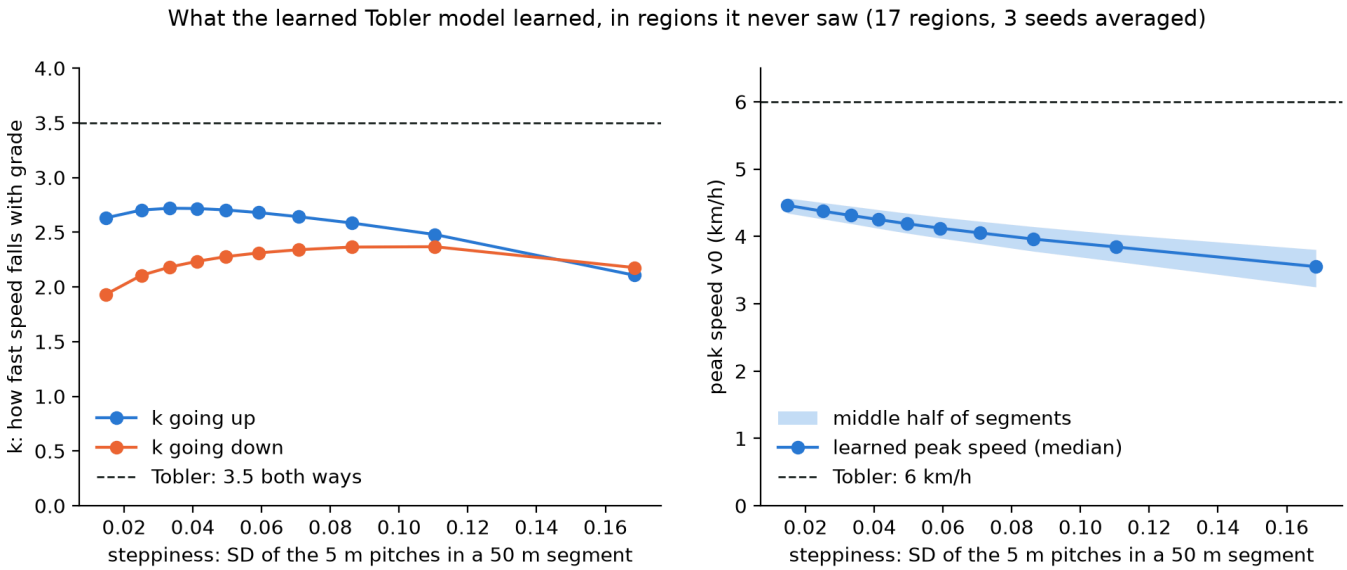

Source: gate2_summary.csv and the paper's Table I. Tags: additive model; terrain: boosted trees; learned Tobler: its own spread.

The one caveat is check 5: learned Tobler's edge comes from its own uncertainty, which the trees did not have in this test.

## What learned Tobler learned

Its constants change with the terrain in sensible ways. These are read in regions it never saw (17 regions, 3 seeds averaged):

- **k**, how fast speed falls as the trail steepens, stays between about 1.9 and 2.7, never Tobler's 3.5.
- **Peak speed** falls from 4.5 km/h on smooth trail to 3.6 km/h on steppy trail (Tobler: 6).
- Going down smooth trail is easy (k down 1.9); steppy trail slows descents too.




Left: k going up and down; right: peak speed. Dashed: Tobler's 3.5 and 6 km/h.

## Stop-rule hand check

The rule that built our dataset misses most short stops; a plain speed rule matches my hand labels better. I labeled 20 GPS moments from White Mountains hikes as stopped or moving (19 usable answers: 7 stops, 12 moving).

| Stop rule | Agrees with me | Stops caught | False stops |
|---|---|---|---|
| Dataset rule: 60 s ending < 10 m away | 8 of 19 | 1 of 7 | 5 |
| Speed rule: < 0.5 km/h | 13 of 19 | 6 of 7 | 5 |
| Either rule | 12 of 19 | 6 of 7 | 6 |

The dataset rule cannot see stops under a minute, and most of mine were 20–40 s. Switching to the speed rule barely moves the White Mountains curve: peak speed 4.23 → 4.13 km/h, k 2.10 → 2.15. My plan: keep the paper's numbers, state this check as a limitation, and use the speed rule in the data release.

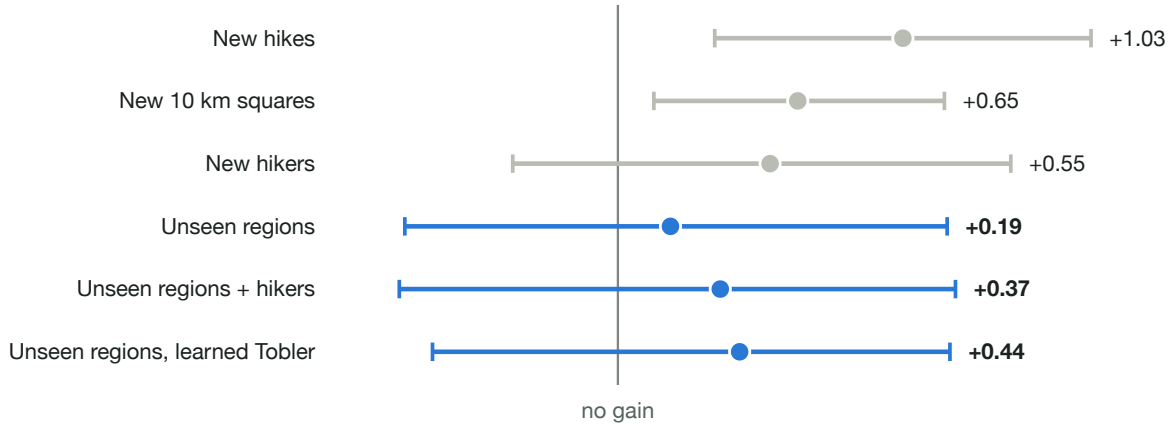
## Imagery Track A: helps where the model has been, not in new regions

Track A used Clay v1.5's ready-made embeddings: 1,024 numbers for every 155 m square of NAIP aerial photos, already computed. Each walking segment took the square under its midpoint. That covers 15 regions:

Clay has Olympic only as 2017 photos at 1 m, and the North Shore is in Canada. In each test fold, PCA shrank the numbers to 16, fitted on that fold's training rows only, and they were added to the Gate 2 models.

**Imagery helps where the model has been (+1.0 on new hikes), not in new regions (+0.2)**

Gain from adding Clay imagery to terrain, in CRPS skill points, with 95% range; 15 regions



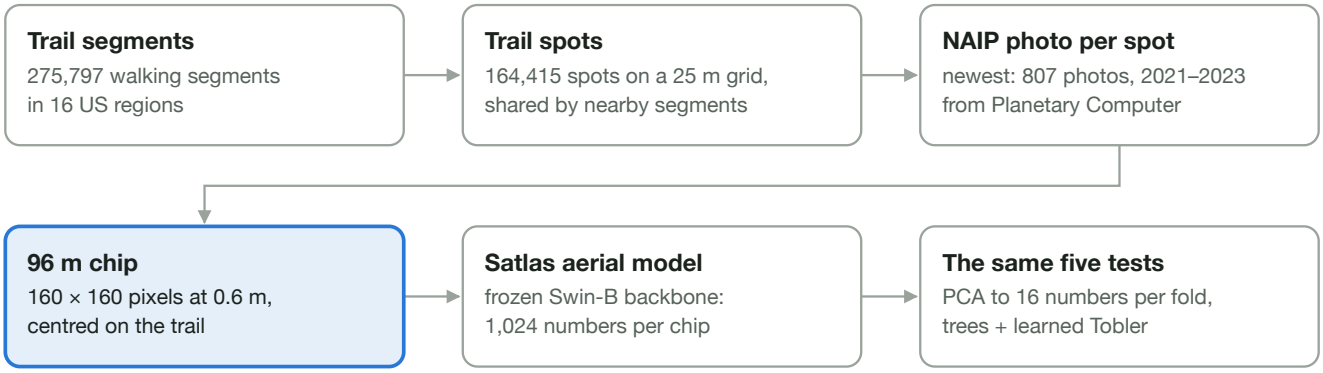

Source: results/t3_imagery/track_a_trees_summary.csv and the learned-Tobler check (decision #125). Trees unless noted.

By the rule I set before running it (a gain above zero in the new-region tests), Track A does not pass. It is the same pattern as the satellite embeddings: the model uses imagery to recognize places it has seen, not to learn trail features that carry over.

## Imagery Track B: photos centred on the trail (running)

Track A's 155 m squares mostly show the landscape around a trail. Track B tests whether a closer look at the trail itself carries over to new regions, with the same tests and the same pass rule.

**Track B looks at the trail itself: a 96 m photo centred on every trail spot**




Decision #126; scripts t3_naip_spots.py and t3_naip_satlas.py.

**Status (6 Nov):** running on the cluster, 16 regions; results expected later today. The first run lost 9 regions because Planetary Computer's download token expires after about 15 minutes. That is fixed, and those regions restarted.

**What the model sees.** Two of the 20 photos I checked by hand, with our matched trail in red: under forest the trail is hidden; in open terrain the path is visible.



White Mountains, forest: the trail (red) is hidden under the canopy.



Front Range, alpine meadow: a faint path runs along the red line.

## Next steps

- **Today or tomorrow:** Track B results, with the same tests and pass rule as Track A.
- **Tue 10 Nov:** the IGARSS submission site opens; the paper moves to the official template.
- **Wed 11 Nov:** decide imagery's place in the paper: a result, or one sentence.
- **16–22 Nov:** the full draft, rewritten in my own words.